Day 116 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Connect Fractional Unit Cubes and Volume

Standards: 6.G.A.2

Learning target: I can connect fractional unit cubes with a prism volume formula.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit. Divide each prism edge by 1/n to count cubes along it. Multiply the counts, then multiply by the volume of one small cube.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 19

Day 116 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Connect Fractional Unit Cubes and Volume

Standards: 6.G.A.2

Learning target: I can connect fractional unit cubes with a prism volume formula.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit. Divide each prism edge by 1/n to count cubes along it. Multiply the counts, then multiply by the volume of one small cube.

Worked example 1

Fill a prism 3/2 × 7/2 × 3 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

1. For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit.

2. Divide each prism edge by 1/n to count cubes along it.

3. Counts along edges: 3, 7, 6.

4. Product 126

5. 126 × 1/8 = 63/4, also lwh.

6. State the result in the form requested: 126 cubes; each 1/8 cubic unit; prism 63/4 cubic units.

Answer: 126 cubes; each 1/8 cubic unit; prism 63/4 cubic units

Worked example 2

A prism measures 11/2 m by 1 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit.

2. Multiply edges: 11/2 × 1 × 1 = 11/2.

3. The edge counts are 11, 2, 2, giving 44 cubes.

4. Each cube is 1/8 m^3

5. 44/8 = 11/2.

Answer: 11/2 m^3; 44 cubes

Day 116 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Fill a prism 3 × 5/2 × 3/2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

02 / Constructed response

Fill a prism 3/2 × 3/2 × 5/2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

03 / Constructed response

Fill a prism 5/2 × 7/2 × 2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

04 / Constructed response

Fill a prism 5/2 × 5/2 × 5/2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

05 / Constructed response

Fill a prism 7/2 × 2 × 3 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

06 / Constructed response

Fill a prism 4 × 3/2 × 2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

07 / Constructed response

Fill a prism 5/3 × 4/3 × 5/3 units with cubes of edge 1/3 unit. How many cubes fit? Use their volumes to verify lwh.

08 / Numeric entry and model reasoning

A prism measures 13/2 m by 3/2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Numeric entry and model reasoning

A prism measures 15/2 m by 2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

10 / Constructed response

Fill a prism 5/3 × 1 × 2 units with cubes of edge 1/3 unit. How many cubes fit? Use their volumes to verify lwh.

Day 116 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Connect Fractional Unit Cubes and Volume

Standards: 6.G.A.2

Learning target: I can connect fractional unit cubes with a prism volume formula.

1. 90 cubes; each 1/8 cubic unit; prism 45/4 cubic units

Counts along edges: 6, 5, 3. Product 90; 90 × 1/8 = 45/4, also lwh.

6.G.A.2 | Constructed response

2. 45 cubes; each 1/8 cubic unit; prism 45/8 cubic units

Counts along edges: 3, 3, 5. Product 45; 45 × 1/8 = 45/8, also lwh.

6.G.A.2 | Constructed response

3. 140 cubes; each 1/8 cubic unit; prism 35/2 cubic units

Counts along edges: 5, 7, 4. Product 140; 140 × 1/8 = 35/2, also lwh.

6.G.A.2 | Constructed response

4. 125 cubes; each 1/8 cubic unit; prism 125/8 cubic units

Counts along edges: 5, 5, 5. Product 125; 125 × 1/8 = 125/8, also lwh.

6.G.A.2 | Constructed response

5. 168 cubes; each 1/8 cubic unit; prism 21 cubic units

Counts along edges: 7, 4, 6. Product 168; 168 × 1/8 = 21, also lwh.

6.G.A.2 | Constructed response

6. 96 cubes; each 1/8 cubic unit; prism 12 cubic units

Counts along edges: 8, 3, 4. Product 96; 96 × 1/8 = 12, also lwh.

6.G.A.2 | Constructed response

7. 100 cubes; each 1/27 cubic unit; prism 100/27 cubic units

Counts along edges: 5, 4, 5. Product 100; 100 × 1/27 = 100/27, also lwh.

6.G.A.2 | Constructed response

8. 39/2 m³; 156 cubes

Multiply edges: 13/2 × 3/2 × 2 = 39/2. The edge counts are 13, 3, 4, giving 156 cubes. Each cube is 1/8 m³; 156/8 = 39/2.

6.G.A.2 | Numeric entry and model reasoning

9. 45/2 m³; 180 cubes

Multiply edges: 15/2 × 2 × 3/2 = 45/2. The edge counts are 15, 4, 3, giving 180 cubes. Each cube is 1/8 m³; 180/8 = 45/2.

6.G.A.2 | Numeric entry and model reasoning

10. 90 cubes; each 1/27 cubic unit; prism 10/3 cubic units

Counts along edges: 5, 3, 6. Product 90; 90 × 1/27 = 10/3, also lwh.

6.G.A.2 | Constructed response